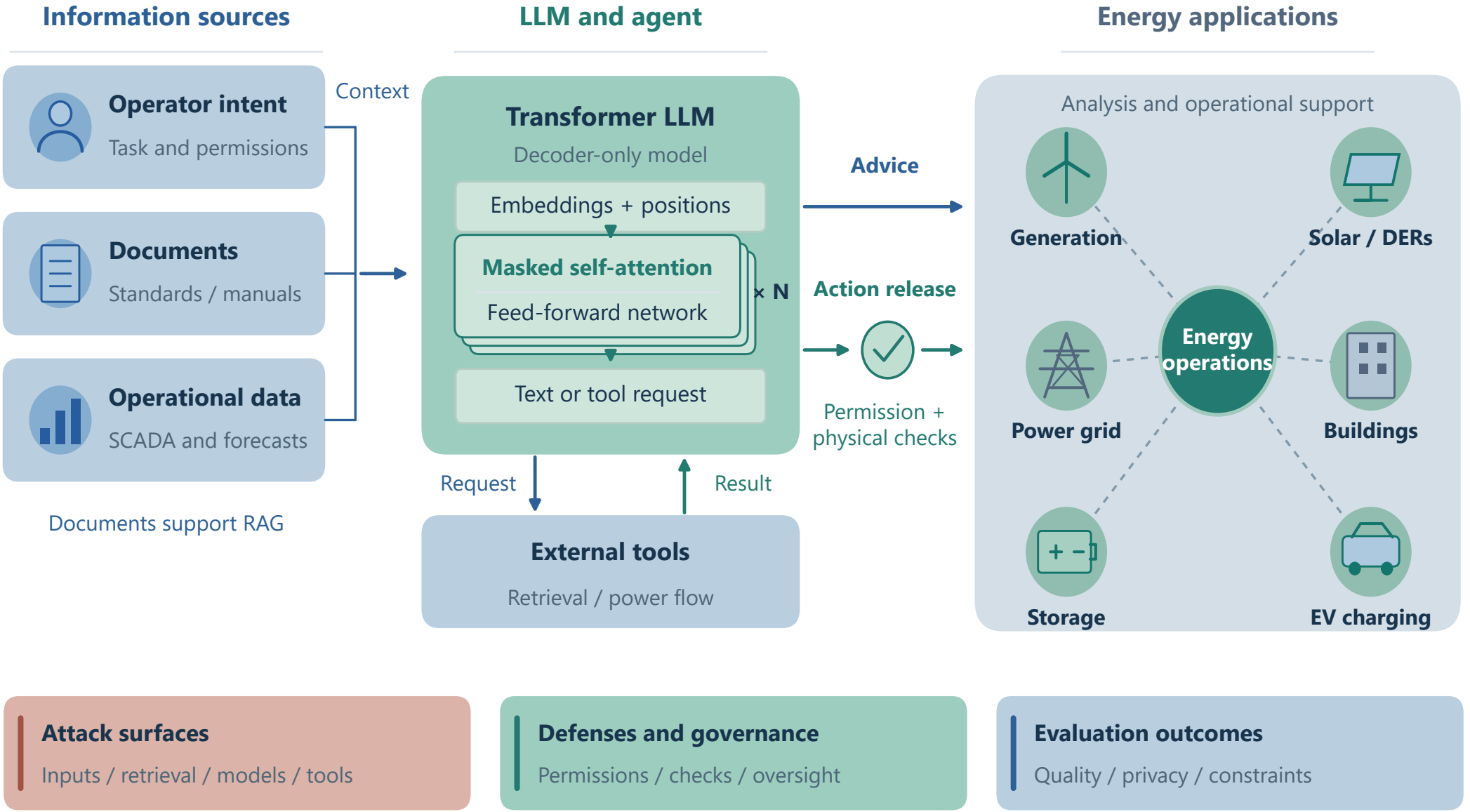
Information sources
LLM and agent
Energy applications
Operator intent
Task and permissions
Documents
Standards / manuals
Operational data
SCADA and forecasts
Documents support RAG
Context
Transformer LLM
Decoder-only model
Embeddings + positions
Masked self-attention
Feed-forward network
× N
Text or tool request
Request
Result
External tools
Retrieval / power flow
Advice
Action release
Permission + physical checks
Analysis and operational support
Generation
Solar / DERs
Power grid
Energy operations
Buildings
Storage
EV charging
Attack surfaces
Inputs / retrieval / models / tools
Defenses and governance
Permissions / checks / oversight
Evaluation outcomes
Quality / privacy / constraints
Arrows: information or release paths. Dashed links: application associations, not electrical connections.